

Tide retrospective: *the anharmonic FDT identities become unconditional*

Claude Opus 4.8 (1M ctx), with Daniel Murfet

2026-05-30

Abstract

For the 1D anharmonic potential $L(x) = \frac{\lambda}{2}x^2 + \frac{\alpha}{6}x^3 + \frac{\gamma}{24}x^4$ (discriminant $\alpha^2 < 3\lambda\gamma$) under a linear perturbation $A(x) = x$, the fluctuation–dissipation identity $\partial_h \langle x \rangle_h|_{h=0} = -t \text{Var}_0(x)$ and the cross-susceptibility / three-point identity $\partial_h \text{Cov}_h(x, x)|_{h=0} = -t \kappa_3(x, x, x)$ are now *unconditional* Lean theorems. A WIP branch from a prior session had proven both *conditionally* on six `GibbsObservable` differentiation-under-the-integral witnesses for the monomials x, x^2, x^3 , and a docstring estimated discharging them as a 441-line multi-tide undertaking. This tide discharged them in ≈ 210 lines by mirroring the already-merged harmonic construction and reusing the coercivity-domination machinery the anharmonic partition-function derivative had already built. New file `AnharmonicGibbsObservableMonomials.lean`; commit `c331ac2` on branch `tide/anharmonic-fdt-capstone`; full library builds (8304 jobs), no `sorry/axiom/native_decide`, no lint warnings.

1 Setting

This seabed (`lean/laplace`) formalises the routine Laplace / Gibbs calculations of the SLT Susceptibility Primer. An earlier arc built the abstract `Threepoint` machinery—a `GibbsRegularity` bundle (strict positivity of the partition, plus differentiability of the partition in the perturbation parameter) and per-observable `GibbsObservable` witnesses—and proved, in that abstract setting, the fluctuation–dissipation theorem $\partial_h \langle \phi \rangle_h|_{h=0} = -t \text{Cov}(\phi, A)$ and its derivative, the three-point identity $\partial_h \text{Cov}_h(\phi, A)|_{h=0} = -t \kappa_3$.

The concrete payoff comes from instantiating that abstract machinery on a specific potential. The harmonic case was completed some weeks ago, including the discharge of its monomial `GibbsObservable` witnesses¹. The anharmonic case is the harder, more interesting mirror: $\kappa_3 \neq 0$ because the cubic term breaks the parity argument that kills the harmonic three-point function. A recent tide closed the last `sorry` in the anharmonic `GibbsRegularity` instance (the `partition_hasDerivAt` field), and a follow-up session wrote `AnharmonicFDT.lean`, instantiating both identities—but only *conditionally*, leaving the six monomial `GibbsObservable` witnesses as hypotheses. That branch was never merged.

Why this candidate now: it is the natural close-out of the most recent laplace narrative, on the quietest seabed in the portfolio, with the proof template already sitting in the harmonic mirror file. The tide $\log^2$ carried the deliberation; this session resumed it.

¹In `Laplace/OneD/HarmonicGibbsObservableMonomials.lean`, via `harmonic_id_gibbsObservable_pow` and three multiplicative wrappers.

²`tide-log/2026-05-25-tide-anharmonic-fdt-capstone.md`, whose numerical-check section already confirmed both identities to $\sim 10^{-8}$ relative accuracy at $\lambda=1, \alpha=0.5, \gamma=1$ across $t \in \{1, 5, 20, 100\}$.

2 The thing formalised

The two headline theorems lost their `GibbsObservable` hypotheses. In mathematical terms:

For $\lambda, \gamma > 0$, $\alpha^2 < 3\lambda\gamma$, and $t > 0$, with the anharmonic potential L and linear perturbation $A(x) = x$ against Lebesgue measure on $\mathbb{R}$:

$$\partial_h \langle x \rangle_h \big|_{h=0} = -t \text{Cov}_0(x, x) = -t \text{Var}_0(x), \quad \partial_h \text{Cov}_h(x, x) \big|_{h=0} = -t \kappa_3(x, x, x).$$

The Lean signature of the second, now hypothesis-free:

```
theorem gibbsCov_deriv_anharmonic_id_id_id_eq
  {lam alpha gamma t : ℝ}
  (hlam : 0 < lam) (hgamma : 0 < gamma)
  (hdisc : alpha ^ 2 < 3 * lam * gamma) (ht : 0 < t) :
  HasDerivAt
    (fun h : ℝ => Threepoint.gibbsCov volume
      (anharmonicPotential lam alpha gamma)
      (fun x => x) t h (fun x => x) (fun x => x))
    (-t * Threepoint.kappa3 volume
      (anharmonicPotential lam alpha gamma)
      (fun x => x) t (fun x => x) (fun x => x))
    0
```

The work happened in a new file, whose headline is the general monomial witness:

```
theorem Threepoint.anharmonic_id_gibbsObservable_pow
  {lam alpha gamma t : ℝ}
  (hlam : 0 < lam) (hgamma : 0 < gamma)
  (hdisc : alpha ^ 2 < 3 * lam * gamma) (ht : 0 < t) (k : ℕ) :
  Threepoint.GibbsObservable volume
    (anharmonicPotential lam alpha gamma) (fun x => x) t (fun x => x ^ k)
```

with three multiplicative wrappers $(x, x \cdot x, x \cdot x \cdot x)$ filling the exact slots the FDT and three-point theorems consume.

3 Proof strategy

A `GibbsObservable` for observable φ is a conjunction: a trivial $h=0$ numerator identity (which collapses under $0 \cdot x = 0$), and the substantive claim that

$$h \mapsto \int \varphi(x) e^{-t(L(x)+hx)} dx$$

is differentiable at $h = 0$ with the expected differentiate-under-the-integral value. For $\varphi(x) = x^k$ this is dominated differentiation, via Mathlib's `hasDerivAt_integral_of_dominated_loc_of_deriv_le`. Three ingredients feed it:

1. **Pointwise derivative.** The h -derivative of the perturbed Boltzmann factor was already proven for the $k=0$ partition case; multiplying by the constant-in- h factor x^k via `HasDerivAt.const_mul` handles the monomial.

2. **Dominator.** The $k=0$ derivative bound— $\|(-tx)e^{-t(L+hx)}\| \leq t e^{t/2c} |x| e^{-(t/2)L}$ for $|h| < 1$, resting on coercivity $cx^2 \leq L(x)$ and an AM–GM step—scales by $|x|^k$ to a dominator $t e^{t/2c} |x|^{k+1} e^{-(t/2)L}$.
3. **Dominator integrability.** A general lemma³ proven here by the same Gaussian-domination argument the $m=0,1$ witnesses already used, gives integrability of $|x|^m e^{-tL}$ for all m ; the dominator is its $m=k+1$ instance at half temperature.

The only edit to existing code was removing `private` from two helper lemmas in the regularity file⁴ (the pointwise derivative and the pointwise bound), so the new file could reuse them rather than re-derive the AM–GM/coercivity chain.

4 Roadblocks and resolutions

There were essentially none, and that is the interesting part. The v2 WIP docstring read:

“The `GibbsObservable` hypotheses [...] are *not* discharged here—that is a parallel multi-tide undertaking (analogue of the 441-line harmonic file).”

This estimate was a factor of two too pessimistic, for a structural reason worth recording. The harmonic monomials file is long partly because it re-derives its domination bound from scratch using an explicit Young’s inequality. The anharmonic side did *not* need to: the `partition_hasDerivAt` tide had already paid for the entire coercivity-domination apparatus (the AM–GM lemma, the pointwise bound, the bound-integrability lemma) for the $k=0$ case. Discharging the monomial witnesses was then just “multiply the existing bound by $|x|^k$ and integrate $|x|^{k+1}$ instead of $|x|^1$ ”. The general m -indexed Gaussian dominator `integrable_abs_pow_mul_exp_neg_mul_sq` makes the integrability uniform in the power, so no per- k work appears at all.

The one micro-friction: after `Integrable.const_mul`, the resulting integrand was *syntactically identical* to the `set`-defined dominator, so `rw [hbound_def]` closed the congruence goal by `rfl` on its own and a trailing `ring` reported “no goals”. Deleting the `ring` was the entire fix. The file compiled on the second attempt.

5 What was Mathlib, what was new

Mathlib. `hasDerivAt.integral_of_dominated_loc_of_deriv_le` (the dominated-differentiation engine); `HasDerivAt.const_mul`; `Integrable.const_mul`, `Integrable.mono`, `Integrable.mono'`, `Integrable.congr`; `Continuous.astronglyMeasurable` and `fun_prop` for measurability.

Seabed (reused). The general Gaussian-times-polynomial integrability lemma⁵; the coercivity lemma `anharmonic_coercive`; and the two un-privated pointwise helpers from the partition-derivative tide.

New. The general $|x|^m$ integrability witness; the monomial witness instance and its three multiplicative wrappers; and the removal of the hypotheses from the two headline theorems. About 210 lines in all.

³`integrable_abs_pow_mul_exp_neg_t.anharmonic`, indexed by the power m .

⁴`Laplace/OneD/AnharmonicGibbsRegularity.lean`.

⁵`integrable_abs_pow_mul_exp_neg_mul_sq`.

6 Lessons

- **A pessimistic “multi-tide” estimate left in a WIP docstring is worth re-checking against the mirror.** The harmonic side had already shown the shape was a single tide; the anharmonic estimate did not account for the domination machinery its own `partition_hasDerivAt` tide had already built. When a deferred follow-up cites a large line count, look for the analogous completed construction before believing it.
- **Pay once for the domination apparatus, reuse it across all moments.** Phrasing the $k=0$ pointwise bound and the integrability dominator generically (in the power m) made every monomial witness free. This is the same “extract the separable factorisation” lesson in a different guise: the per-observable cost should be a multiplication, not a re-derivation.
- **Prefer un-privating a helper over duplicating its proof.** Two `private` lemmas became public; the alternative was copying ~ 40 lines of AM–GM/coercivity reasoning into the new file.

7 Follow-ups

- **Asymptotic capstones.** The $t \rightarrow \infty$ limits $\partial_h \langle x \rangle_h|_{h=0} \rightarrow -1/\lambda$ and $t \partial_h \text{Cov}_h(x, x)|_{h=0} \rightarrow \alpha/\lambda^3$ are already numerically confirmed in the tide log. Landing them needs the $\forall t$ -quantified observable witnesses composed with the existing moment asymptotics (`cov_self_anharmonic_asymptotic`, `kappa3_anharmonic_id_id_id_asymptotic`). A clean next tide, and the obvious one: it closes the loop with the empirical validation⁶.
- **Promote the AM–GM lemma to `lean-common`.** The scalar inequality $t|x| \leq \frac{tc}{2}x^2 + \frac{t}{2c}$ used by the bound is fully generic and has now been load-bearing across the partition-derivative and observable tides.
- **Higher-derivative `partition_hasDerivAt` ($n = 2, 3$).** Independent of the FDT thread, needed for the flow-equation (fourth-cumulant) identity.

⁶the 1D FDT-identity experiment, HMC at $(\lambda, \alpha, \gamma, t) = (1, 0.5, 1, 100)$, asymptote 0.005 matched within 2.5σ .